Core Compression Map for Collective Excitation Systems

A Structural Framework for Photonic Slow Light, Phonon Dispersion and Polariton Hybridization

Name:
Youfu Wang (王有福)

ORCID:
0009-0005-5998-6024

Abstract

Many collective excitation systems exhibit similar structural behavior near dispersion boundaries. Examples include slow light in photonic structures, phonon dispersion flattening in crystals, and hybrid mode formation in polariton systems.

Although these phenomena arise in different physical platforms, their structural evolution shows strong similarities when approaching critical dispersion regimes.

In this work we introduce a structural representation called the Core Compression Map, defined in the phase plane

$$(\mathcal J,\Delta_{\text{stab}})$$

where $\mathcal J$ represents interaction compression and Δ_{stab} denotes stability margin.

The framework identifies four structural regimes:

- geometry basin (G)
- gap ridge (B)
- resonance wall (R)
- exit zone (X)

Three representative physical systems serve as standard windows anchoring the unified trajectory

$$G \rightarrow B \rightarrow R \rightarrow X .$$

Photonic slow-light systems occupy the geometric regime, phonon dispersion flattening corresponds to the threshold regime, and polariton hybrid modes approach the hybridization regime.

Experimental dispersion relations can be mapped onto the Core Map through group velocity and dispersion curvature. The framework provides a structural description that may help compare collective excitation phenomena across different physical systems.

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1 Introduction

Collective excitation phenomena occur in many areas of physics, including photonics, condensed matter physics, and light–matter interaction systems.

Several well-known examples include:

- slow light in photonic crystal waveguides
- phonon dispersion flattening in crystalline lattices
- exciton–polariton hybrid modes in semiconductor microcavities

Despite the different microscopic origins of these systems, they often display similar structural behavior near dispersion boundaries. Typical signatures include:

- reduction of group velocity
- flattening of dispersion relations
- enhanced interaction or coupling
- hybridization between modes

Traditionally, these phenomena are analyzed within the specific theoretical framework of each field. Photonic slow light is typically described using photonic band theory, phonon dispersion using lattice dynamics, and polariton systems using light–matter coupling models.

However, when viewed from a structural perspective, these phenomena exhibit remarkable similarities. This suggests that different systems approaching dispersion boundaries may follow a common structural evolution.

In this work we introduce a structural representation called the Core Compression Map, which provides a unified geometric framework for describing these phenomena.

Rather than focusing on microscopic interactions, the Core Map describes systems using two macroscopic structural variables:

- compression strength
- stability margin

Within this representation, different physical systems can be located on a common phase diagram and their structural evolution can be represented as trajectories within this diagram.

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2 Λ-Field Representation

To describe structural compression we introduce a scalar field

$$\Lambda(x,t)$$

which represents the degree of structural compression within a system.

The Λ-field does not correspond to a single physical observable. Instead it represents a collective structural parameter capturing effects such as

- dispersion flattening
- interaction enhancement
- mode coupling

In different systems the Λ-field can be interpreted differently:

system	Λ-field interpretation
photonic systems	band flattening
phonon systems	lattice dispersion compression
polariton systems	hybridization strength

Thus the Λ-field provides a unified way to represent structural compression across different physical platforms.

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3 Unified Action

The dynamics of the Λ-field are described by the action

$$S=\int d^4x \left[\frac12(\partial_\mu \Lambda)^2 - V(\Lambda) \right]$$

where $V(\Lambda)$ is a structural potential that determines the stability of the system.

The stability margin is defined as

$$\Delta_{\text{stab}} = V''(\Lambda).$$

Stable configurations satisfy

$$\Delta_{\text{stab}} > 0 .$$

As the system approaches a structural boundary,

$$\Delta_{\text{stab}} \rightarrow 0 .$$

This condition signals the approach to a critical regime in the Core Compression Map.

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4 Core Compression Map

System states are represented in a two-dimensional phase diagram defined by

$$(\mathcal J,\Delta_{\text{stab}})$$

where

$$\mathcal J$$

represents compression strength.

For systems with measurable dispersion relations we define

$$\mathcal J = \frac{1}{v_g}$$

where v_g is the group velocity.

The stability margin is defined as

$$\Delta_{\text{stab}}
\sim
\frac{d^2 \omega}{dk^2}$$

which measures the curvature of the dispersion relation.

The Core Compression Map naturally divides into four structural regions:

Region	Name	Structural Meaning
G	Geometry Basin	stable propagation
B	Gap Ridge	threshold competition
R	Resonance Wall	hybridization regime

X Exit Zone instability

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5 Structural Trajectories

As system parameters vary, the system evolves along trajectories

$\Gamma(s)$

in the Core Compression Map.

Three basic trajectory classes appear:

Type-G

Stable geometric regime:

$G \rightarrow G$

Type-B

Threshold compression regime:

$G \rightarrow B$

Type-R

Hybridization regime:

$G \rightarrow B \rightarrow R \rightarrow X$

These trajectories represent increasing structural compression and interaction strength.

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6 Unified Charge

The internal structural load of the system is described by the Unified Charge

$$\mathcal C =
\mathcal E_\Lambda
+$$

$$
\zeta_1 \mathcal Q
+
\zeta_2 \mathcal M .
$$

The three components represent different structural contributions:

component	meaning
E_Λ	structural energy
Q	stability quota
M	hybridization load

Charge evolution follows

$$
\frac{d\mathcal C}{ds}
=
\left(\frac{d\mathcal C}{ds}\right)_{\text{int}}
+
\left(\frac{d\mathcal C}{ds}\right)_{\text{tr}} .
$$

The first term represents internal redistribution, while the second describes branch transitions.

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7 State Charge

The overall structural phase of the system is described by the State Charge

$$
\mathcal U
=
\mathcal C
+
\kappa_1 \Delta_{\text{stab}}
+
\kappa_2 \mathcal J .
$$

Different regions of the Core Map correspond to different structural states:

Region	State
G	stable storage
B	compression competition
R	hybridization
X	instability

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8 Cross-System Mapping

Three representative systems illustrate the Core Compression Map.

Photonic slow light

Slow-light systems in photonic crystal waveguides show strong reduction in group velocity near band edges.

These systems occupy the region

$$G \rightarrow B$$

in the Core Map.

Phonon dispersion

Phonon dispersion flattening near Brillouin zone boundaries corresponds to the

$$B$$

region of the map.

Polariton hybrid modes

Exciton–polariton systems exhibit strong light–matter coupling and mode hybridization, placing them near

$$R .$$

Together these systems trace the structural trajectory

$$G \rightarrow B \rightarrow R .$$

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9 Discussion

The Core Compression Map suggests that many collective excitation systems may follow a common structural evolution when approaching dispersion boundaries.

Rather than treating photonic, phononic, and polaritonic systems separately, the map provides a geometric framework for comparing them.

Possible extensions include:

- magnon dispersion in magnetic systems
- superconducting gap dynamics

These systems may test the boundaries of the structural map.

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10 Conclusion

We introduced the Core Compression Map, a structural phase representation defined in the plane

$$(\mathcal J,\Delta_{\text{stab}}).$$

Three representative systems—photonic slow light, phonon dispersion, and polariton hybrid modes—serve as standard windows anchoring the structural trajectory

$$G \rightarrow B \rightarrow R \rightarrow X .$$

The framework introduces unified charge and state charge to describe structural load and global organization.

The Core Compression Map may provide a structural framework for comparing collective excitation phenomena across multiple physical platforms.
